MUHMMAD SOHAIB

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Education

FAST - National University of Computer & Emerging Sciences Lahore, PK
BS Data Science 2023-2027

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Systems, Software Engineering

Experience

Speedy Drive Car Rental, Dubai, UAE Remote
AI Engineer (Voice AI and Automation) Sept 2025 – Feb 2026

- Deployed a production **Voice AI Agent** handling end-to-end car rental sales calls — availability, dynamic pricing, insurance, delivery, and booking confirmation — with **zero human intervention**
- Engineered real-time **backend tool functions** for live availability lookup, **SQL**-based pricing, total cost aggregation, and **TTS-formatted** automated booking confirmation emails
- Implemented **Vapi** tool-calling architecture for mid-call structured data injection, eliminating pricing and availability hallucinations via strict tool orchestration
- Developed a real-time **analytics dashboard** (TypeScript) tracking call outcomes, booking statuses, success rates, and revenue across Daily/Weekly/Monthly views
- Engineered **agent system prompt** covering multi-department routing, dynamic tool orchestration, fallback handling, TTS-compliant formatting, and conditional booking logic

Projects

AI Virtual Try-On
TypeScript, Fastify, BullMQ, Redis, REST APIs

- Architected an **async backend pipeline** using Fastify with multipart endpoints accepting product image URLs and user-uploaded images for AI try-on processing
- Implemented **BullMQ + Redis job queue** with full lifecycle tracking (queued → processing → completed) and real-time status polling via REST
- Added UUID-based file storage, input validation, and structured error handling for reliable async task processing at scale

Business Insights and Sales Forecasting Tool
Python, FastAPI, TypeScript, PostgreSQL, React, TensorFlow, Prophet, LangGraph, vector stores, HuggingFace

- **System Architecture:** Architected a full-stack decision intelligence platform splitting concerns between a TypeScript-based Node.js operational backend for inventory and procurement, and a Python FastAPI microservice dedicated to machine learning inference and stateful agentic workflows.
- **Transactional Integrity & Auditing:** Implemented database concurrency controls utilizing Prisma transactions and database row-level locking to prevent race conditions during inventory updates, backed by custom middleware logging detailed write-audit records with JSON diffs and IP tracking.
- **Iterative Forecasting Pipeline:** Deployed a time-series forecasting engine utilizing Meta’s Prophet model for seasonal baselines and a production Random Forest Regressor using iterative lag-feature recursion for multi-step revenue projections, integrated with MAE, RMSE, and MAPE performance monitoring.

- **Deep Learning Churn Engine:** Developed a customer churn prediction model using a TensorFlow Keras Deep Neural Network with Batch Normalization, Dropout, and SMOTE oversampling to correct historical transaction imbalance, achieving an AUC-ROC of 0.82 against Scikit-learn baselines.
- **Multi-Agent RAG:** Designed a LangGraph-coordinated multi-agent chatbot utilizing Llama 3.1 and 3.3 models for dynamic planning, database tool execution, and local Hugging Face vector retrieval, secured by a query-sanitization layer that blocks malicious SQL inputs.

Smart Movie Recommendation System

Python, NLP, scikit-learn, EDA, TMDB API

- Performed **EDA** analyzing genre distributions, popularity trends, and feature correlations to guide model design
- NLP-based **content-based recommendation engine** consolidating genres, cast, crew, and keywords into a unified tags vector for cosine similarity matching
- Integrated **TMDB API** to render posters and metadata for top-5 recommendations in an interactive UI

Airline Customer Satisfaction Predictor

Python, scikit-learn, Random Forest, Logistic Regression, KNN, EDA

- Conducted **EDA** identifying key satisfaction drivers: seat comfort, service quality, flight distance, and customer loyalty patterns
- Multi-model **binary classification pipeline**; Random Forest achieved best accuracy, precision, recall, and F1 over Logistic Regression and KNN baselines

Skills & Interests

Voice AI and Agents: Voice AI Agent Development, Tool-Calling Architecture, Prompt Engineering, Agent System Design, TTS Optimization, LLM Orchestration, Workflow Automation

Machine Learning and AI: scikit-learn, XGBoost, Random Forest, NLP, Feature Engineering, Content-Based Filtering, TimeSeries Forecasting, EDA, Data Visualization

Backend and APIs: TypeScript, Node.js, Fastify, REST APIs, BullMQ, Redis, Async Pipelines, Multipart Handling, Third-Party API Integration

Data and Databases: Python, SQL (relational schema design, lead storage, CRM pipelines), Pandas, Data Cleaning and Preprocessing

Tools: Git, GitHub, Postman, n8n, make, Retell, PowerBI,